

Patent wars?

As reported by ZedNet, at a 3D printing conference in New York, John Hornick, from the law firm Finnegan headquartered at Washington DC said that, similar to the wars with smartphones, the 3D printing wars would begin soon if the market capitalisation figure is big enough to protect intellectual property at any cost.

Paul Anand, CEO, Biotz supports this saying, "With all the top shots like Intel, Microsoft and HP from 2D printing industry entering into the field of 3D printing the competition is going to be high. This would not only increase the use of technologies but also help them improve a lot by bringing down the price of the 3D printers to the consumer level."

Printing technologies with a twist

Certain 3D printing technologies have evolved with specific changes based on environment. Let us take a look at the most popular of these:

Carbon fibre. Carbon fibre can usually be incorporated into objects only by hand, so working with them is time consuming and proves expensive too. 3D printers use carbon fibre material in the form of composites, which cuts down waste by providing the exact physical length needed without any requirement to trim.

Light-directed electrophoretic deposition (EPD). EPD is an old technology that is unable to allow selective depositions at certain spots. A combination of photoconductive electrodes with a DC electric field allows spot deposition to produce accurate composites in a process called light-directed EPD. EPD being a faster deposition technique, the addition of light to it allows the creation of accurate void areas, thus helping the engineers to come up with all types of designs in a shorter span of time.

The interesting point in this technology is that it uses laser-cut aluminium mask, which is likely to be replaced soon by a digitally-projected mask that would make it a completely automated deposition system. Being a resin-type 3D printing system, it provides better finish to the object being printed.

Biomimicry. Biomimicry is a new science that studies nature's models and then uses these designs and processes to solve human problems. 3D printers can be a great boon in making products similar to those found in nature. The best

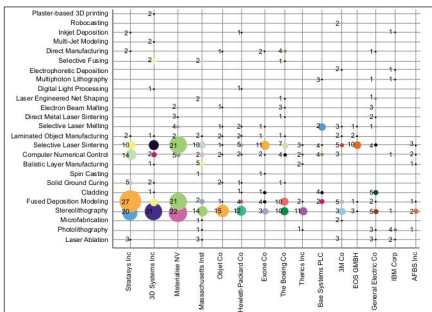


Fig. 1: Technologies used by top 15 brands (Courtesy: Patent INSIGHT Pro)

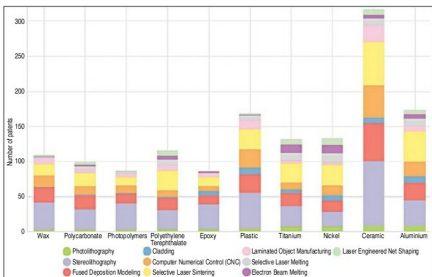


Fig. 2: Key materials used in different 3D printing technology (Courtesy: Patent INSIGHT Pro)

thing about this technology is that the final products can get different colours without any extra cartridge attachment to the 3D printers. This lets us create new prototypes of different colours and shapes without any extra tools and cost.

Sensor and data analysis in 3D printing. Sensors are the key source for inspection in data analysis. They play an important role in improving the additive manufacturing process in 3D printing. If we take the example of selective layer sintering (SLS), the sensor would allow the engineers to identify the optimal coupling between the powders and the

laser by helping them to analyse the right amount of energy transferred into the powder layer. It helps to prevent overheating during the process.

New 3D printers in the market

The 3D printing industry can be seen as a growing industry with developments being made at every stage. No wonder newer technologies are emerging. But there is requirement of new 3D printers that can be used for multiple platforms with multiple filaments.

There is a 3D printer MarkOne, created by a start up named MarkForged,